ITA06020 MACHINE LEARNING

Experiment 8 – Implementation of Linear Regression

Aim:

To implement the Linear Regression algorithm in Python to predict house prices based on different housing features using a real-world dataset.

Dataset: California Housing Dataset (real-world dataset available in Scikit-learn).

Procedure

1. Import the required Python libraries.
2. Load the California Housing dataset.
3. Separate input features and target values.
4. Split the dataset into training and testing data.
5. Create a Linear Regression model.
6. Train the model using training data.
7. Predict house prices using testing data.
8. Calculate and display the Mean Squared Error and R^2 score.

Python Program

```
from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

data = fetch_california_housing()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Linear Regression")

print("Actual Price:", y_test[0], "($100,000 units)")
```

```
print("Predicted Price:", y_pred[0], "($100,000 units)")

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", round(mse, 4))

print("R2 Score:", round(r2, 4))
```

Output:

```
Linear Regression
Actual Price: 0.477 ($100,000 units)
Predicted Price: 0.7191228416021644 ($100,000 units)
Mean Squared Error: 0.5559
R2 Score: 0.5758
```

Result

Thus, the Linear Regression algorithm was successfully implemented using Python and the California Housing dataset to predict house prices.